

# UK AIR

## Air Information Resource

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## PM<sub>2.5</sub> Targets (PERT and AMCT)

The [Environmental Targets \(fine particulate matter\) \(England\) Regulations 2023](#) set two PM<sub>2.5</sub> targets for England to be met by the end of 2040. Interim targets for 2028 were published in the [Environmental Improvement Plan 2023](#). Information on how progress towards these targets is assessed is provided below; an overview of the targets themselves can be found [here](#) and evidence relating to development of the targets can be found [here](#).

Progress towards both targets is assessed using measurements of PM<sub>2.5</sub> concentration from the Automatic Urban and Rural Network (AURN) [monitoring network](#), which is being expanded to support this assessment. Information on how PM<sub>2.5</sub> is measured can be found [here](#).

## The population exposure reduction target

The population exposure reduction target (PERT) sets a minimum percentage reduction in the population exposure to be achieved between the base year (2018) and target deadline (2040). Population exposure is represented by the Population Exposure Indicator (PEI), calculated using measurements of annual mean concentration at representative monitoring sites. The PERT monitoring sites are spread geographically across England; reflect population density; and are located in areas where concentrations are typical of where most people in that region live. Monitoring sites characterised as ‘urban background’ (and, in some instances, ‘suburban background’) meet these criteria. Other considerations, such as representing areas of deprivation, are also taken into account when establishing new sites. The PEI is a three-year average to reduce interannual variability due to factors such as weather and other temporary external events and provide a more stable trend. Population exposure has an established link with health impacts, so a decrease in the PEI represents a reduction in the risk to the nation’s health from PM<sub>2.5</sub>.

The [calculation approach](#) incorporates a statistical method recommended by experts to account for the large change in the number of PM<sub>2.5</sub> monitoring sites expected as a result of the ongoing expansion of the network. A direct comparison of the PEI in 2040 with 2018 may otherwise be influenced by network changes. The solution involves calculating yearly increments in PEI using the same cohort of monitoring sites, then summing the increments and comparing this with the base year PEI. The [Air Quality Expert Group \(AQEG\)](#) has been engaged in the development of this approach.

## PERT progress

As of 2024, population exposure has reduced by 25% compared to 2018.

The Population Exposure Indicator (PEI) is shown for each year in the table below. A third column shows the change in PEI compared to the previous year. A fourth column then shows the cumulative change, i.e., the total change since 2018 (i.e. the PEI base year). The fifth column then shows the total percentage change in PEI since 2018.

The base year (2018) population exposure index, PEI<sub>base</sub>, has been calculated to be 10.09 µg m<sup>-3</sup>. In 2024, the PEI is 7.56 µg m<sup>-3</sup>, which represents a yearly change in PEI of -0.3 µg m<sup>-3</sup>. This represents a 25% reduction since 2018 (-2.53 µg m<sup>-3</sup>).

More information on how these statistics are calculated can be found [here](#).

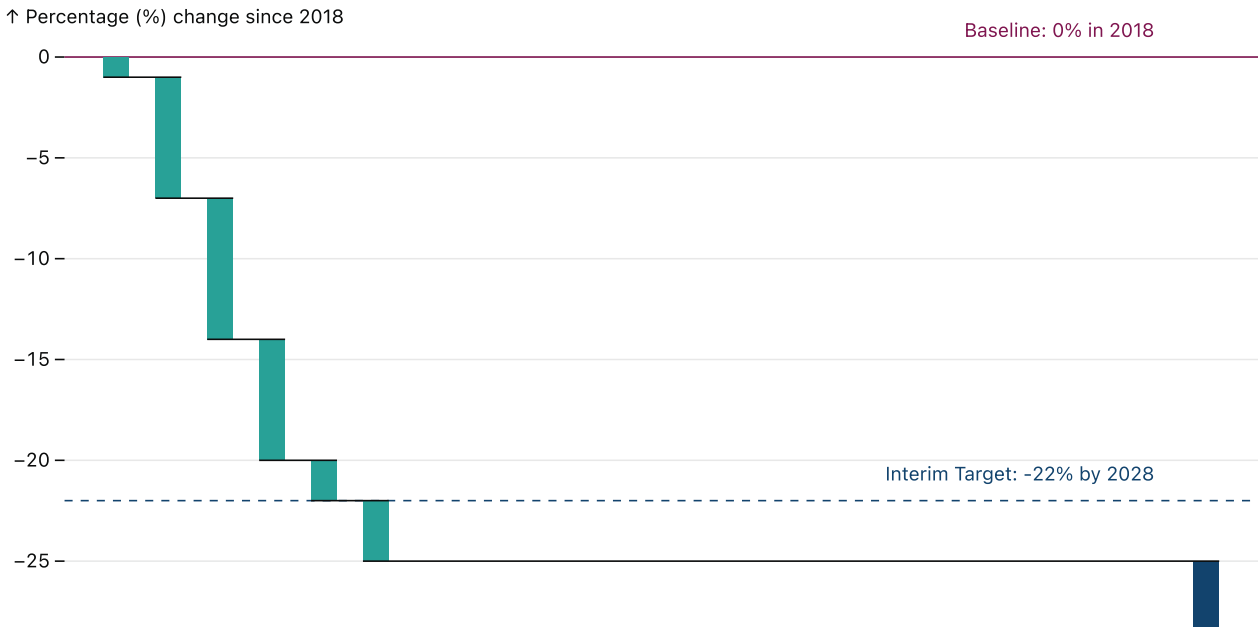
| Year | PEI<br>( $\mu\text{g m}^{-3}$ ) | Yearly Change<br>in PEI<br>( $\mu\text{g m}^{-3}$ ) | Total Change in PEI<br>since 2018<br>( $\mu\text{g m}^{-3}$ ) | Total Change in PEI<br>since 2018<br>(%) |
|------|---------------------------------|-----------------------------------------------------|---------------------------------------------------------------|------------------------------------------|
| 2019 | 10.02                           | -0.07                                               | -0.07                                                         | -1%                                      |
| 2020 | 9.41                            | -0.61                                               | -0.68                                                         | -7%                                      |
| 2021 | 8.68                            | -0.73                                               | -1.41                                                         | -14%                                     |
| 2022 | 8.11                            | -0.57                                               | -1.98                                                         | -20%                                     |
| 2023 | 7.86                            | -0.25                                               | -2.23                                                         | -22%                                     |
| 2024 | 7.56                            | -0.30                                               | -2.53                                                         | -25%                                     |

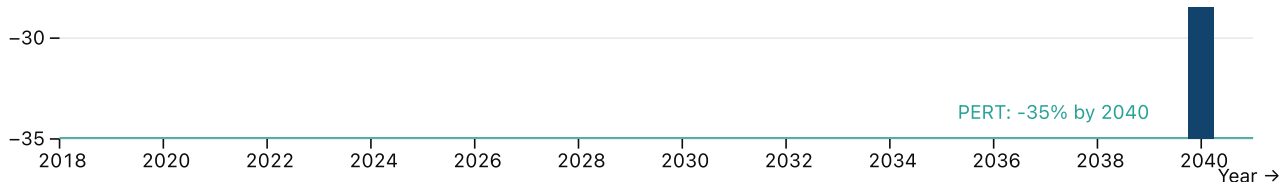
Progress is also shown in the waterfall chart. Turquoise bars represent decreases, orange bars represent any increases, and the dark blue bar shows the remaining reduction in PEI needed to achieve the PERT. The lines in the graph show the 2028 interim target of 22% and the legally binding target of 35% for 2040.

The graph shows that the PM<sub>2.5</sub> concentrations people in England are exposed to is decreasing over time. As the PERT metric is based on a three-year average there is less variation than with annual concentrations. Analysis suggests the large decrease from 2020 onwards is most likely due to decreases in both UK and European secondary PM<sub>2.5</sub> (some of the PM<sub>2.5</sub> we experience originates from outside the UK). In 2024, average exposure was 25% less than 2018, which is greater than the interim target of 22% to be met by 2028.

Interactive Graphics

Hover over each of the segments of the waterfall chart to see the change in PEI that year and the cumulative change in PEI since 2018. Hovering over the dark blue bar will show the difference between the cumulative change in PEI and the PERT.





## The annual mean concentration target

The annual mean concentration target (AMCT) sets a maximum annual mean concentration of  $10 \mu\text{g m}^{-3}$  which all English AURN sites (i.e., [near-source as well as background](#)) need to meet by the target deadline of 2040. The annual mean concentration indicates the concentration at a particular location, and varies from year-to-year due to a range of factors such as weather conditions, as well as pollutant emissions. A reduction in the maximum annual mean concentration measured and number of locations above the AMCT shows that concentrations are reducing in the areas where they are the highest.

## AMCT progress

In 2024, the highest measured annual mean concentration was  $11 \mu\text{g m}^{-3}$  and 99% of English AURN sites which met the required data capture threshold (78 individual AURN sites) were equal to or below the  $10 \mu\text{g m}^{-3}$  annual mean concentration target. In 2009 the highest annual average concentration was  $18 \mu\text{g m}^{-3}$  and 11% of sites were equal to or below  $10 \mu\text{g m}^{-3}$  (3 individual AURN sites). This is a  $7 \mu\text{g m}^{-3}$  reduction in the highest concentrations over 15 years. The calculation method for the AMCT can be found [here](#).

The bubble chart below shows the distribution of annual PM<sub>2.5</sub> for different English AURN sites in comparison with the AMCT. Each individual scatter marker corresponds to an individual year and integer-rounded annual PM<sub>2.5</sub> concentration, with the size of the marker corresponding to the number of AMCT sites it represents. The lines indicate the 2028 interim target of  $12 \mu\text{g m}^{-3}$ , and the 2040 AMCT of  $10 \mu\text{g m}^{-3}$ . Markers are coloured to show whether the site is above the interim target (orange), between the interim and AMCT (dark blue), or below the AMCT (turquoise).

Trends for each site vary, but in general concentrations have been decreasing at the majority of sites. The large decrease in the concentrations at most sites over the past five years is most likely due to decreases in both UK and European secondary PM<sub>2.5</sub>. In 2023, no sites were above the 2028 interim target and one site ([London Marylebone Road, MY1](#)) was above the long-term target.

The number of PM<sub>2.5</sub> monitoring sites is being increased as part of a network expansion programme so this will affect these statistics over the next few years.

**Data Archive (<https://uk-air.defra.gov.uk/data/>)**

**PM<sub>2.5</sub> Targets (PERT and AMCT) (<https://uk-air.defra.gov.uk/pm25targets/progress>)**

Methodology for calculating progress towards the PM<sub>2.5</sub> targets (<https://uk-air.defra.gov.uk/pm25targets/calculation>)

Data Selector ([https://uk-air.defra.gov.uk/data/data\\_selector](https://uk-air.defra.gov.uk/data/data_selector))

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UK Particle Monitoring Programme (<https://uk-air.defra.gov.uk/data/particle-data>)

Marylebone Road, Non-Automatic Data (<https://uk-air.defra.gov.uk/data/maryleboneroad>)

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